Input

Target RTL

RTL

+

Approximate Unit Library

...

Configuration-Aware Graph

Co-Evolution

Candidate states

- Unevaluated
- Predicted PPA / SSIM
- Delay calibrated
- Measured

y = (Area, Power, Delay, SSIM)

Candidate generation
NSGA-III

Surrogate model

GNN

Area / Power | Delay | SSIM

max

Path readout

PPA / SSIM prediction

DC / SSIM
Real evaluation

Feedback selection

Local calibration
Delay only

Reference feedback

Output

Verified Pareto Front

Power (↓)

Area (↓)

Representative Designs

...

Final DSE

Final search

Selected designs

PPA + SSIM verification

BSDS inputs

Verified results